

**Course Grade = Rounded(Unrounded Exam Grade+Bonus)**

| Points | Unrounded Exam Grade | Rounded Exam Grade |
|--------|----------------------|--------------------|
| 0      | 1.000                | 1                  |
| 1      | 1.091                | 1                  |
| 2      | 1.182                | 1.25               |
| 3      | 1.273                | 1.25               |
| 4      | 1.364                | 1.25               |
| 5      | 1.455                | 1.5                |
| 6      | 1.545                | 1.5                |
| 7      | 1.636                | 1.75               |
| 8      | 1.727                | 1.75               |
| 9      | 1.818                | 1.75               |
| 10     | 1.909                | 2                  |
| 11     | 2.000                | 2                  |
| 12     | 2.091                | 2                  |
| 13     | 2.182                | 2.25               |
| 14     | 2.273                | 2.25               |
| 15     | 2.364                | 2.25               |
| 16     | 2.455                | 2.5                |
| 17     | 2.545                | 2.5                |
| 18     | 2.636                | 2.75               |
| 19     | 2.727                | 2.75               |
| 20     | 2.818                | 2.75               |
| 21     | 2.909                | 3                  |
| 22     | 3.000                | 3                  |
| 23     | 3.091                | 3                  |
| 24     | 3.182                | 3.25               |
| 25     | 3.273                | 3.25               |
| 26     | 3.364                | 3.25               |
| 27     | 3.455                | 3.5                |
| 28     | 3.545                | 3.5                |
| 29     | 3.636                | 3.75               |
| 30     | 3.727                | 3.75               |
| 31     | 3.818                | 3.75               |
| 32     | 3.909                | 4                  |
| 33     | 4.000                | 4                  |
| 34     | 4.077                | 4                  |
| 35     | 4.154                | 4.25               |
| 36     | 4.231                | 4.25               |
| 37     | 4.308                | 4.25               |
| 38     | 4.385                | 4.5                |
| 39     | 4.462                | 4.5                |
| 40     | 4.538                | 4.5                |
| 41     | 4.615                | 4.5                |
| 42     | 4.692                | 4.75               |

|    |       |      |
|----|-------|------|
| 43 | 4.769 | 4.75 |
| 44 | 4.846 | 4.75 |
| 45 | 4.923 | 5    |
| 46 | 5.000 | 5    |
| 47 | 5.077 | 5    |
| 48 | 5.154 | 5.25 |
| 49 | 5.231 | 5.25 |
| 50 | 5.308 | 5.25 |
| 51 | 5.385 | 5.5  |
| 52 | 5.462 | 5.5  |
| 53 | 5.538 | 5.5  |
| 54 | 5.615 | 5.5  |
| 55 | 5.692 | 5.75 |
| 56 | 5.769 | 5.75 |
| 57 | 5.846 | 5.75 |
| 58 | 5.923 | 6    |
| 59 | 6.000 | 6    |
| 60 | 6.000 | 6    |
| 61 | 6.000 | 6    |
| 62 | 6.000 | 6    |
| 63 | 6.000 | 6    |
| 64 | 6.000 | 6    |

# Grading Scheme *Analysis I: eine Variable*

Grading Collective

February 2024

## 1 Kästchenaufgaben

### 1.1 K1

Two points if correct, one point if only  $+C$  was forgotten and 0 points otherwise.

### 1.2 K2

Two points if correct, one point if only  $+C$  was forgotten and 0 points otherwise.

### 1.3 K3

Two points if correct, one point if only the error term is wrong and 0 points otherwise.

## 2 Beweise aus der Vorlesung

### 2.1 B1

D Stating explicitly that we can use uniform continuity since we have a continuous function on a compact domain.

Z Take an interval decomposition such that the distance between successors is smaller than the  $\delta$  in uniform continuity.

F1 Define  $c_k$  and  $d_k$ , together with mentioning why they exist.

F2 Argue why  $d_k - c_k < \varepsilon$ .

S1 Define the two step functions  $l$  and  $u$ .

S2 Explicitly remark  $l \leq f \leq u$ .

C Show that the  $\int_a^b u - l < \varepsilon \cdot C$  for some constant  $C$  and conclude.

## 2.2 B2

### 2.2.1 $\alpha < 1$

SUP write the correct Def. of lim sup or follow directly that  $x_n \rightarrow \alpha$  for  $n \rightarrow \infty$

q correctly define a q, or at least a q s.t.  $\alpha < q < 1$  and at least get to second next step of  $|a_k| < q^k$

k if step inbetween and correct use of N and k in  $|a_k| < q^k$  (for all  $k \geq N$ )  
correct as well: existence of N  $\Rightarrow x_N < q$

geo from convergence of geometric series and the Majorant criterium follows  
convergence, still get the point if they do not mention the names but  
explicitly state as  $q < 1$  series  $q^k$  convergence and therefore series of  $|a_k|$   
as well

### 2.2.2 $\alpha > 1$

acc Limsup is accumulation point  $\Rightarrow$  subsequence convergence to same point  
s.t.  $\lim_{k \rightarrow \infty} \sqrt[n_k]{|a_{nk}|} > 1$  (still get point if everything defined except  
for naming accumulation point, correct definition of subsequence and lim  
important)

N exist N s.t.  $\sqrt[n_k]{|a_{nk}|} > 1$  for all  $k \geq N$

Z sequence does not converge to zero  $\Rightarrow$  series does not converge

## 3 Anwendungen der Theorie

### 3.1 A1

#### 3.1.1 A1 a) + b)

Part a)

I  $f'$  is increasing (either prove this or use the theorem from lectures)

R inequality and reverse case  $y < x$ . It is sufficient to mention the reverse  
case with an idea why it works. (still get point if case  $x = y$  is missing)

Part b)

I  $f'$  is increasing

C  $f$  is convex (either prove this or use the theorem from lectures). This  
point is only given if there is a good attempt at the first part.

Proof using twice differentiable:

D  $f$  is convex (complete correct proof needed to get 1 point)

### 3.1.2 A1 c)

#### Case 1 - Fundamental Theorem of Calculus

- A) separation of cases  $h > 0$ ,  $h < 0$ . If the case  $h < 0$  is not detailed, expecting at least a sentence summarizing the idea
- B) mentioning the FTC
- C) using monotonicity correctly and concluding (only mentioning monotonicity is not sufficient as it was proved in the above subquestions)

#### Case 2 - Equivalent characterization of convexity

- A) separation of cases  $h > 0$ ,  $h < 0$ . If the case  $h < 0$  is not detailed, expecting at least a sentence summarizing the idea
- B) stating the inequality characterizing slopes for three points on a convex function
- C) using differentiability of  $f$  correctly to take the limit on one side of the inequality and conclude

#### Case 3 - Mean Value Theorem

- A) separation of cases  $h > 0$ ,  $h < 0$ . If the case  $h < 0$  is not detailed, expecting at least a sentence summarizing the idea
- B) mentioning and stating the MVT (even if the application of it is wrong)
- C) using monotonicity of  $f'$  correctly to conclude

#### Case 4 - None of these types of proofs could be identified

- A) separation of cases  $h > 0$ ,  $h < 0$ . If the case  $h < 0$  is not detailed, expecting at least a sentence summarizing the idea

can still be given, regardless of the correctness of the proof.

## 3.2 A2

### 3.2.1 A2 a)

#### Case 1 - Alternating Series

- A) Observe that  $x_n = \sum_{k=0}^n (-\alpha)^k$  (Point is only given if they write it as a Series)
- B) Proof A) by induction
- C) Argue that the limit is an infinite geometric series with  $|q| < 1$  or that  $\alpha^n$  is a monotonically decreasing zero-sequence and Leibnitz' Criterion gives convergence

D) Recall the limit of a geometric sequence

E) Correctly compute the limit (up to minor computational mistakes)

Rem: If the student writes down a slightly different series but argues everything else correctly he can get all points except (B).

### Case 2 - Split into monotone sequences

A) Attempt to split  $x_n$  into  $x_{2n}$  and  $x_{2n+1}$

B) Proof that  $x_{2n}$  and  $x_{2n+1}$  are monotone

C) Proof that  $x_{2n}$  and  $x_{2n+1}$  are bounded (only gives points if the student distinguishes the two sequences somewhere in his solution)

D) Fixed-point formula using  $x_{2(n+1)} = 1 - \alpha + \alpha^2 x_{2n}$

E) Correctly compute both limits and see that they are equal

Rem: If at any point the student writes down  $x_{2(n+1)} = 1 - \alpha + \alpha^2 x_{2n}$  I gave them the point for considering splitting the sequence. In some cases this was **very** generous.

Rem: If the student doesn't compute both limits (for example just using  $A = 1 - \alpha A$ ), then he does not get the point for  $D$ . The choice of letter is a bit arbitrary but the point is that they can't be sure that the sequence even converges.

### 3.2.2 A2 b)

**Case  $\alpha = 1$**

*Solution 1*

A) Compute the first terms and remark that the sequence is alternating.

B) Prove the general alternating scheme (i.e. with recurrence or with the closed form) and conclude.

*Solution 2*

A) Recall from question a)/prove the closed form for  $x_n$ .

B) State that the geometric series does not converge for general term  $|a_n| \geq 1$

**Case  $\alpha > 1$**

*Solution 1*

C) Remark that the sequence alternates between  $x_{2k} \geq 1$  and  $x_{2k+1} < 0$ .

D) Prove the above inequalities and conclude.

*Solution 2*

- C) Recall from question a)/prove the closed form for  $x_n$ .
- D.1) State that the geometric serie does not converge for general term  $|a_n| \geq 1$  and conclude.
- D.2) Alternatively, can state criterion on convergence of series (Cauchy, Leibniz, etc) and conclude.

*Solution 3*

- C) Remark that  $x_n$  has a diverging subsequence and find it.
- D) Prove that it is indeed a subsequence, prove divergence and conclude.

*Solution 4*

- C) Remark that  $x_n$  is a polynom which leading term is  $\alpha^n$ , which will eventually diverge.
- D) Prove it and conclude.

*Solution 5*

- C) Express the distance  $|x_{n+1} - x_n|$  in terms of  $\alpha$ .
- D) Prove the distance  $|x_{n+1} - x_n|$  diverges and so  $x_n$  is non-cauchy.

# Aufgaben und Lösungsvorschlag

## Multiple Choice

**MC1 [2 Punkte]** Welche der folgenden Mengen ist nicht offen?

*Which among the following sets is not open?*

- (A)  $(0, 1) \cup (2, \infty)$
- (B)  $(0, 1) \cup [1, 2)$
- (C)  $[0, 1) \cup (1, 2)$
- (D)  $[1, 2) \cup (0, 3)$

**Lösung:**

C.

**MC2 [2 Punkte]** Welche der folgenden Aussagen ist korrekt?

*Which of the following statements is correct?*

- (A) Jede Cauchy-Folge ist monoton.  
*Every Cauchy sequence is monotone.*
- (B) Jede monotone Folge ist beschränkt.  
*Every monotone sequence is bounded.*
- (C) Jede beschränkte und monotone Folge ist eine Cauchy-Folge.  
*Every bounded and monotone sequence is a Cauchy sequence.*
- (D) Jede beschränkte Folge konvergiert.  
*Every bounded sequence is convergent.*

**Lösung:**

C

**MC3 [2 Punkte]** Sei  $f : [0, \infty) \rightarrow [0, \infty)$  eine stetige Funktion so dass  $f(0) = 0$ . Welche der folgenden Implikationen ist immer korrekt?

*Let  $f : [0, \infty) \rightarrow [0, \infty)$  be a continuous function with  $f(0) = 0$ . Which of the following implications is always correct?*

- (A) Ist  $f$  gleichmässig stetig, so ist  $f$  beschränkt.  
*If  $f$  is uniformly continuous, then  $f$  is bounded.*
- (B) Ist  $f$  nicht beschränkt, so ist  $f$  surjektiv.  
*If  $f$  is unbounded, then  $f$  is surjective.*
- (C) Ist  $f$  monoton, so ist  $f$  injektiv.  
*If  $f$  is monotone, then  $f$  is injective.*
- (D) Ist  $f$  Lipschitz-stetig, so ist  $f$  differenzierbar.  
*If  $f$  is Lipschitz continuous, then  $f$  is differentiable.*

**Lösung:**

**B****MC4 [2 Punkte]** Finden Sie das Minimum auf  $\mathbb{R}$  der Funktion*Find the minimum on  $\mathbb{R}$  of the function*

$$f(x) = x^4 - 2x^2 + 5.$$

(A) 2

(B) 4

(C) 5

(D) 3

**Lösung:**

**B:** The function goes to infinity at  $\pm\infty$  and the derivative is  $4x^3 - 4x = 4x(x^2 - 1)$  that vanishes at  $x = 0, 1, -1$ . By a simple check, the minimum is attained at 1 and  $-1$  and it is equal to

$$1 - 2 + 5 = 4.$$

**MC5 [2 Punkte]** Berechnen Sie den Konvergenzradius der folgenden Potenzreihe:*Compute the radius of convergence of the following power series:*

$$\sum_{n=1}^{\infty} \frac{(2n)^{n-1} + 3^n}{3^n + n^n} x^n.$$

(A)  $\frac{1}{2}$ (B)  $\frac{2}{3}$ (C)  $\frac{3}{2}$ 

(D) 1

**Lösung:**

**A:** The dominant term in the numerator is  $(2n)^{n-1}$ , while in the denominator is  $n^n$ . So

$$\sqrt[n]{\frac{(2n)^{n-1} + 3^n}{3^n + n^n}} \sim \sqrt[n]{\frac{(2n)^{n-1}}{n^n}} = \sqrt[n]{\frac{2^{n-1}}{n}} \rightarrow 2,$$

so the radius of convergence is  $\frac{1}{2}$ .

**MC6 [2 Punkte]** Berechnen Sie das Integral*Compute the integral*

$$\int_0^{\pi^2} \cos(\sqrt{x}) dx.$$

(A) 0

(B) 2

(C)  $-4$

(D)  $-1$

**Lösung:**

**C:** First, by the change of variable  $s = \sqrt{x}$  (so that  $x = s^2$  and  $dx = 2sds$ ) we get

$$\int_0^{\pi^2} \cos(\sqrt{x}) dx = 2 \int_0^{\pi} \cos(s)s ds.$$

Then, using integration by parts,

$$\int_0^{\pi} \cos(s)s ds = [\sin(s)s]_0^{\pi} - \int_0^{\pi} \sin(s) ds = 0 - [-\cos(s)]_0^{\pi} = -2.$$

Hence

$$\int_0^{\sqrt{\pi}} \cos(\sqrt{x}) dx = -4.$$

**MC7 [2 Punkte]** Welche der folgenden Folgen hat weder einen endlichen Grenzwert noch divergiert sie (also konvergiert nicht)?

*Which of the following sequences has no finite limit nor an improper limit (and thus does not converge)?*

(A)  $x_n = \frac{n^2-1}{\log(n)-3n^2}$

(B)  $x_n = \frac{n^3-n^2 \sin(n)}{2n^3+(-1)^n n^2}$

(C)  $x_n = \frac{n+\log(n)}{\log^2(n)+(-1)^n n}$

(D)  $x_n = \frac{n^3-\sqrt{n}}{3n^2-(-1)^n n}$

**Lösung:**

**C**

**MC8 [2 Punkte]** Seien  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  zwei Funktionen. Welche der folgenden Aussagen ist immer korrekt?

*Let  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  be two functions. Which of the following statements is always correct?*

(A) Ist  $g \circ f$  injektiv, so ist  $f$  injektiv.

*If  $g \circ f$  is injective, then  $f$  is injective.*

(B) Ist  $g$  surjektiv, so ist  $g \circ f$  surjektiv.

*If  $g$  is surjective, then  $g \circ f$  is surjective.*

(C) Ist  $g \circ f$  injektiv, so ist  $g$  injektiv.

*If  $g \circ f$  is injective, then  $g$  is injective.*

(D) Ist  $g \circ f$  surjektiv, so ist  $f$  surjektiv.

*If  $g \circ f$  is surjective, then  $f$  is surjective.*

**Lösung:**

A.

**MC9 [2 Punkte]** Berechnen Sie den Grenzwert*Compute the limit*

$$\lim_{x \rightarrow \infty} \sqrt{x} \left( \sqrt{x+1} - \sqrt{x} \right).$$

- (A) 0  
(B)  $\infty$   
(C) 2  
(D)  $\frac{1}{2}$

**Lösung:****D:** We have to compute the limit as  $x \rightarrow \infty$  of

$$\sqrt{x} \left( \sqrt{x} \sqrt{1 + \frac{1}{x}} - \sqrt{x} \right) = x \left( \sqrt{1 + \frac{1}{x}} - 1 \right).$$

Replacing  $\frac{1}{x} = s$ , this is equivalent to

$$\lim_{s \rightarrow 0} \frac{\sqrt{1+s} - 1}{s}.$$

Since  $\sqrt{1+s} = 1 + \frac{1}{2}s + O(s^2)$ , we get

$$\lim_{s \rightarrow 0} \frac{\sqrt{1+s} - 1}{s} = \lim_{s \rightarrow 0} \frac{\frac{1}{2}s + O(s^2)}{s} = \frac{1}{2}.$$

**Lösung:****C:** We split the series in two and analyze the different cases: consider first

$$\sum_{n=1}^{\infty} \frac{n^{\alpha}}{n^2 + n^{2\alpha}}.$$

For  $\alpha \leq 1$  this behaves like  $\sum_{n=1}^{\infty} \frac{n^{\alpha}}{n^2}$  so it converges for  $\alpha < 1$ . For  $\alpha > 1$  it behaves like  $\sum_{n=1}^{\infty} \frac{n^{\alpha}}{n^{2\alpha}} = \sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}$  that always converges for  $\alpha > 1$ . So the first part of the series converges for every  $\alpha \neq 1$ .

For the second series we have

$$\sum_{n=1}^{\infty} \frac{n}{n^2 + n^{2\alpha}}.$$

that only converges for  $\alpha > 1$ .In conclusion, the whole series converges only for  $\alpha > 1$ .**MC10 [2 Punkte]** Welches der folgenden uneigentlichen Integrale konvergiert?*Which of the following improper integrals converges?*

- (A)  $\int_{-\infty}^{\infty} \frac{\sin(x^2)}{1+x^2} dx$   
 (B)  $\int_{-\infty}^{\infty} \frac{e^x}{1+e^{|x|}} dx$   
 (C)  $\int_0^{\infty} \frac{1+\cos(x)}{x^2} dx$   
 (D)  $\int_0^{\infty} \frac{|\sin(x)|}{x} dx$

**Lösung:**

A

**MC11 [4 Punkte]** Betrachten Sie die DGL erster Ordnung

*Consider the first order ODE*

$$u' = -\sin(x)u + \sin(x), \quad u(0) = 0.$$

Finden Sie  $u(\pi)$ .

*Find  $u(\pi)$ .*

- (A)  $1 - e$   
 (B)  $1 + e^{-1}$   
 (C)  $1 - e^{-2}$   
 (D)  $e^{-1}$

**Lösung:**

C: We write

$$\frac{u'}{1-u} = \sin(x) \implies \log(1-u) = \cos(x) + C,$$

thus  $u(x) = 1 - Ce^{\cos(x)}$ . Imposing the initial condition we get  $C = 1/e$ , hence

$$u(\pi) = 1 - \frac{1}{e}e^{\cos(\pi)} = 1 - e^{-2}.$$

**MC12 [4 Punkte]** Betrachten Sie die DGL zweiter Ordnung

*Consider the second order ODE*

$$u'' + u' - 2u = -2x, \quad u(0) = \frac{3}{2}, \quad u'(0) = 2.$$

Finden Sie  $u(1)$ .

*Find  $u(1)$ .*

- (A)  $e + 3/2$   
 (B)  $e + 2$   
 (C)  $e^{-2}$   
 (D)  $e^{-2} + 1$

**Lösung:**

**A:** The solutions to the homogeneous problem are  $e^x$  and  $e^{-2x}$ . Also, a particular solution is given by  $x + \frac{1}{2}$ . So the general solution is given by

$$u(x) = Ae^x + Be^{-2x} + x + \frac{1}{2}.$$

Imposing the boundary conditions we get  $A = 1$  and  $B = 0$ , hence

$$u(x) = e^x + x + \frac{1}{2}.$$

In particular,  $u(1) = e + \frac{3}{2}$ .

## Kästchenaufgaben

**K1 [2 Punkte]** Berechnen Sie das unbestimmte Integral

*Compute the indefinite integral*

$$\int \frac{3}{(1+x^2)(4+x^2)} dx.$$

**Lösung:**

Factor the integrand as

$$\frac{3}{(1+x^2)(4+x^2)} = \frac{1}{1+x^2} - \frac{1}{4+x^2}.$$

The primitive of  $\frac{1}{1+x^2}$  is  $\arctan(x)$ . For the second term, we change variables: choosing  $u = x/2$  we have

$$\int \frac{1}{4+x^2} dx = \frac{1}{4} \int \frac{1}{1+(x/2)^2} dx = \frac{1}{2} \int \frac{1}{1+u^2} du = \frac{1}{2} \arctan(u) + C = \frac{1}{2} \arctan(x/2) + C.$$

Hence

$$\int \frac{3}{(1+x^2)(4+x^2)} dx = \arctan(x) - \frac{1}{2} \arctan(x/2) + C.$$

**K2 [2 Punkte]** Berechnen Sie das unbestimmte Integral

*Compute the indefinite integral*

$$\int \frac{1}{\cos(x) + 1} dx.$$

**Lösung:**

With the half-angle substitution  $u = \tan(x/2)$  we get

$$\int \frac{1}{\cos(x) + 1} dx = \int \frac{1}{\frac{1-u^2}{1+u^2} + 1} \frac{2}{1+u^2} du = \int du = u + C = \tan(x/2) + C.$$

**K3 [2 Punkte]** Berechnen Sie das Taylorpolynom bei 0 bis zum dritten Grad der Funktion

*Compute the Taylor polynomial in 0, up to order 3, of the function*

$$\cos^2(x) - \cos(x^2)e^{x^2}.$$

Verwenden Sie die Gross- $O$  Landau Notation um das Restglied anzugeben.

*Use the big- $O$  Landau notation to indicate the remainder.*

**Lösung:**

Since  $\cos(y) = 1 - \frac{y^2}{2} + O(y^4)$ , we have

$$\begin{aligned}\cos(x^2) - \cos(x^2)e^x &= \left(1 - \frac{x^2}{2} + O(x^4)\right)^2 - \left(1 + O(x^4)\right)\left(1 + x^2 + O(x^4)\right) \\ &= 1 - 2\frac{x^2}{2} - 1 - x^2 + O(x^4) = -2x^2 + O(x^4).\end{aligned}$$

## Beweise aus der Vorlesung

**B1 [7 Punkte]** Zeigen Sie, dass jede stetige Funktion  $f : [a, b] \rightarrow \mathbb{R}$  Riemann-integrierbar ist.

*Show that every continuous function  $f : [a, b] \rightarrow \mathbb{R}$  is Riemann integrable.*

**Lösung:**

Let  $f : [a, b] \rightarrow \mathbb{R}$  be continuous, and  $\varepsilon > 0$ . Since any continuous function on a compact interval is uniformly continuous,  $f$  is uniformly continuous and so there exists  $\delta > 0$  such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

holds for all  $x, y \in [a, b]$ .

Now, let  $a = x_0 < x_1 < \dots < x_n = b$  be a decomposition of  $[a, b]$  such that  $x_k - x_{k-1} < \delta$  for all  $k$ , and for each  $k \in \{1, \dots, n\}$  define

$$c_k = \min\{f(x) \mid x_{k-1} \leq x \leq x_k\} \quad \text{and} \quad d_k = \max\{f(x) \mid x_{k-1} \leq x \leq x_k\}$$

where we used that any continuous function attains its maximum and minimum on a compact interval.

We note that, for all  $k \in \{1, \dots, n\}$ , the inequality  $d_k - c_k < \varepsilon$  holds. Indeed, if  $y_k, z_k \in [x_{k-1}, x_k]$  as points of minimum and maximum, namely  $f(y_k) = c_k$  and  $f(z_k) = d_k$ , then because  $|y_k - z_k| \leq x_k - x_{k-1} < \delta$ , the uniform continuity implies that  $d_k - c_k < \varepsilon$ .

We now define step functions  $\ell, u$  by

$$\ell(x) = \begin{cases} c_k & \text{if } x \in [x_{k-1}, x_k) \\ c_n & \text{if } x = b \end{cases} \quad \text{and} \quad u(x) = \begin{cases} d_k & \text{if } x \in [x_{k-1}, x_k) \\ d_n & \text{if } x = b \end{cases}$$

for  $x \in [a, b]$ . Since  $c_k \leq d_k$ , we see that  $\ell \leq f \leq u$ . Also, because  $d_k - c_k < \varepsilon$ , it follows that

$$\int_a^b (u - \ell) dx = \sum_{k=1}^n (d_k - c_k)(x_k - x_{k-1}) < \varepsilon \sum_{k=1}^n (x_k - x_{k-1}) = \varepsilon(b - a).$$

Since  $\varepsilon > 0$  is arbitrary,  $f$  is integrable.

**B2 [7 Punkte]** Zeigen Sie: Ist  $(a_n)_{n=0}^\infty$  eine Folge reeller Zahlen und

*Show: If  $(a_n)_{n=0}^\infty$  is a sequence of real numbers, and*

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \in \mathbb{R} \cup \{\infty\}.$$

Dann

Then

$$\alpha < 1 \implies \sum_{n=0}^{\infty} a_n \text{ konvergiert absolut / converges absolutely}$$

und

and

$$\alpha > 1 \implies \sum_{n=0}^{\infty} a_n \text{ konvergiert nicht. / does not converge.}$$

## Lösung:

Suppose  $\alpha < 1$ , and define  $q = \frac{1+\alpha}{2} \in (\alpha, 1)$ . Recalling that by definition

$$\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sup_{k \geq n} \sqrt[k]{|a_k|},$$

we have that  $x_n = \sup_{k \geq n} \sqrt[k]{|a_k|} \rightarrow \alpha$  as  $n \rightarrow \infty$ . Hence, since  $\alpha < q$ , there exists  $N \in \mathbb{N}$  such that

$$x_N = \sup_{k \geq N} \sqrt[k]{|a_k|} < q.$$

This yields  $|a_k| < q^k$  for all  $k \geq N$ , so the series  $\sum_{k=1}^{\infty} |a_k|$  converges according to Majorant criterion and the convergence of the geometric series.

If  $\alpha > 1$  holds, since the limsup is an accumulation point, there exists a subsequence  $(a_{n_k})_{k=0}^{\infty}$  such that  $\lim_{k \rightarrow \infty} \sqrt[n_k]{|a_{n_k}|} > 1$ . In particular, there exists  $N \in \mathbb{N}$  such that

$$\sqrt[n_k]{|a_{n_k}|} > 1 \quad \text{for all } k \geq N,$$

or equivalently,  $|a_{n_k}| > 1$  for all  $k \geq N$ . In particular, the sequence  $(a_n)_{n=0}^{\infty}$  does not converge to zero and therefore,  $\sum_{n=0}^{\infty} a_n$  does not converge.

## Anwendungen der Theorie

**A1 [7 Punkte]** Sei  $f : \mathbb{R} \rightarrow \mathbb{R}$  eine differenzierbare Funktion.

*Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a differentiable function.*

a) **[2 Punkte]** Angenommen dass  $f$  konvex ist, zeigen Sie, dass

*Assuming that  $f$  is convex, prove that*

$$(f'(y) - f'(x))(y - x) \geq 0 \quad \forall x, y \in \mathbb{R}.$$

b) **[2 Punkte]** Umgekehrt, angenommen dass

*Vice versa, assume that*

$$(f'(y) - f'(x))(y - x) \geq 0 \quad \forall x, y \in \mathbb{R},$$

zeigen Sie, dass  $f$  konvex ist.

*prove that  $f$  is convex.*

c) **[3 Punkte]** Angenommen  $f$  ist konvex, zeigen Sie dass

*Assuming that  $f$  is convex, prove that*

$$f(x + h) \geq f(x) + f'(x)h \quad \forall x, h \in \mathbb{R}.$$

*Erinnerung:* Eine Funktion  $f : \mathbb{R} \rightarrow \mathbb{R}$  ist konvex wenn für alle  $a, b \in \mathbb{R}$  so dass  $a < b$  und alle  $t \in (0, 1)$  gilt dass

*A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is convex if for all  $a, b \in \mathbb{R}$  with  $a < b$  and all  $t \in (0, 1)$ , it holds that*

$$f((1 - t)a + tb) \leq (1 - t)f(a) + tf(b).$$

### Lösung:

a) Since  $f$  is convex,  $f'$  is increasing. So, if  $x < y$  then  $f'(x) \leq f'(y)$  and the inequality follows. The case  $x > y$  is analogous.

b) One can note that the inequality

$$(f'(y) - f'(x))(y - x) \geq 0 \quad \forall x, y \in \mathbb{R}.$$

implies that  $f'$  is increasing, thus the result.

*Alternative argument for b) if  $f$  is twice differentiable (still worth points).* Take  $y = x + h$  with  $h > 0$ . Then the inequality gives

$$(f'(x + h) - f'(x))h \geq 0.$$

Dividing by  $h^2$  and letting  $h \rightarrow 0$ , this gives

$$\frac{f'(x + h) - f'(x)}{h} \geq 0 \quad \implies \quad f''(x) \geq 0 \quad \forall x \in \mathbb{R}.$$

Thus,  $f$  is convex.

- c) Consider first the case  $h > 0$ . By the fundamental theorem of calculus and the monotonicity of  $f'$  (note that  $f'$  is monotone, thus integrable) we get

$$f(x+h) - f(x) = \int_x^{x+h} f'(s) ds \geq f'(x) \int_x^{x+h} ds = f'(x)h,$$

as desired. If  $h < 0$  we have

$$f(x) - f(x+h) = \int_{x+h}^x f'(s) ds \leq f'(x) \int_{x+h}^x ds = -f'(x)h,$$

which again gives the result.

**Alternative Argument 1.** If  $h = 0$ , the inequality trivially holds. If  $h > 0$ , consider the closed interval  $[x, x+h]$ . By the Mean Value Theorem, there exists  $\xi \in (x, x+h)$  such that

$$f'(\xi) = \frac{f(x+h) - f(x)}{h}.$$

Since  $f$  is convex, its derivative is increasing and therefore  $x < \xi$  implies

$$f'(x) \leq f'(\xi) = \frac{f(x+h) - f(x)}{h}.$$

Multiplying with  $h > 0$  and subsequently adding  $f(x)$  on both sides then gives the result. For  $h < 0$ , we consider the interval  $[x+h, x]$ . We argue as above and note that multiplying with  $h < 0$  corrects the inequality going the wrong way.

**Alternative Argument 2.** For  $h = 0$ , the inequality trivially holds. For  $h > 0$ , pick any  $y \in (x, x+h)$ . The equivalent condition for convexity then reads

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(x+h) - f(y)}{x+h-y}.$$

We then take the limit  $y \rightarrow x$  on both sides (note that the limit on the LHS exists since  $f$  is differentiable) and obtain

$$f'(x) \leq \frac{f(x+h) - f(x)}{h}.$$

Multiplying with  $h > 0$  and subsequently adding  $f(x)$  on both sides gives the result.

If  $h < 0$ , pick any  $y \in (x+h, x)$  and note that multiplying with  $h < 0$  accounts for the inequality going the other way.

**A2 [9 Punkte]** Für  $\alpha > 0$ , betrachten Sie die Folge

*Given  $\alpha > 0$ , consider the sequence*

$$x_0 = 1, \quad x_{n+1} = 1 - \alpha x_n \quad \text{for } n \geq 0.$$

- a) **[5 Punkte]** Zeigen Sie, dass die Folge für  $\alpha \in (0, 1)$  konvergiert und bestimmen Sie den Grenzwert.

*Prove that for  $\alpha \in (0, 1)$ , the sequence converges and identify the limit.*

b) [4 Punkte] Zeigen Sie, dass die Folge für  $\alpha \geq 1$  nicht konvergiert.

*Prove that for  $\alpha \geq 1$ , the sequence does not converge.*

*Hinweis: Betrachten Sie die Fälle  $\alpha = 1$  und  $\alpha > 1$  separat.*

*Consider the cases  $\alpha = 1$  and  $\alpha > 1$  separately.*

### Lösung:

a) If a limit  $L \in \mathbb{R}$  exists, then it satisfies

$$L = 1 - \alpha L \implies L = \frac{1}{1 + \alpha}.$$

Using this relation, we have that  $1 = L(1 + \alpha)$ , hence

$$x_{n+1} = 1 - \alpha x_n = L(1 + \alpha) - \alpha x_n \implies x_{n+1} - L = \alpha(L - x_n).$$

Taking the absolute value, we get

$$|x_{n+1} - L| = \alpha |x_n - L| \implies |x_n - L| = \alpha^n |x_0 - L|.$$

If  $\alpha < 1$ , the relation above implies that  $|x_n - L| \rightarrow 0$  exponentially fast.

b) If  $\alpha > 1$ , the relation above implies that  $|x_n - L| \rightarrow \infty$  exponentially fast, so the sequence does not converge.

If  $\alpha = 1$  then we can immediately check that

$$x_0 = 1, \quad x_1 = 1 - x_0 = 0, \quad x_2 = 1 - x_1 = 1, \quad x_3 = 0, \dots,$$

and more in general

$$x_{2n} = 1, \quad x_{2n+1} = 0,$$

so the sequence does not converge.

**Alternative argument 1.** If  $\alpha < 1$  one can see by induction that

$$x_1 < x_3 < x_5 < \dots < L < \dots < x_4 < x_2 < x_0.$$

Hence the sequence  $x_{2n}$  is decreasing and converges to a limit  $L_1 \geq L$ , while the sequence  $x_{2n+1}$  is increasing and converges to a limit  $L_2 \leq L$ .

From the relation  $x_{n+1} = 1 - \alpha x_n$  we get

$$x_{n+2} = 1 - \alpha x_{n+1} = 1 - \alpha(1 - \alpha x_n) = 1 - \alpha + \alpha^2 x_n.$$

So, the limit of  $x_{2n}$  satisfies

$$L_1 = 1 - \alpha + \alpha^2 L_1 \implies L_1 = \frac{1 - \alpha}{1 - \alpha^2} = \frac{1}{1 + \alpha} = L,$$

and analogously for  $L_2$ . This proves that  $x_n \rightarrow L$ .

If  $\alpha = 1$ , we argue as before.

If  $\alpha > 1$  one can see by induction that

$$\dots < x_5 < x_3 < x_1 < L < x_0 < x_2 < x_4 < \dots$$

so the sequence cannot converge.

**Alternative Argument for a).** We first prove by induction that

$$x_n = \sum_{i=0}^n (-1)^i \alpha^i$$

- **Base Case:**

$$x_0 = 1 = (-1)^0 \alpha^0$$

- **Induction Step:**

$$\begin{aligned} x_{n+1} &= 1 - \alpha x_n \\ &= 1 - \alpha \sum_{i=0}^n (-1)^i \alpha^i \\ &= 1 + \sum_{i=0}^n (-1)^{i+1} \alpha^{i+1} \\ &= 1 + \sum_{i=1}^{n+1} (-1)^i \alpha^i \\ &= \sum_{i=0}^{n+1} (-1)^i \alpha^i \end{aligned}$$

Now  $\lim_{n \rightarrow \infty} x_n = \sum_{i=0}^{\infty} (-1)^i \alpha^i$ . From the lecture, we know that for  $\alpha \in (0, 1)$  that  $a_i := \alpha^i$  is a monotonically decreasing zero-sequence. Thus Cauchy Leibnitz from the lecture implies the sum converges. Finally, we can compute

$$\lim_{n \rightarrow \infty} x_n = \sum_{i=0}^{\infty} (-\alpha)^i = \frac{1}{1 - (-\alpha)} = \frac{1}{1 + \alpha}$$

since  $(-\alpha)^i$  is a geometric sequence with  $|a_i| < 1$ .